

Program: Diploma in Fire Technology and Safety	
Course Code: 6469	Course Title: Safety Engineering Lab
Semester: 6	Credits: 1.5
Course Category: Program Elective	
Periods per week: 3 (L:0, T:0, P:3)	Periods per semester: 45

Course Objectives:

- To provide hands-on experience in safety procedures and hazard analysis.
- To familiarize students with safety equipment and risk assessment techniques.
- To develop skills in accident investigation and emergency response planning.

Course Outcomes:

On completion of the course, the student will be able to:

CO n	Description	Duration (Hours)	Cognitive Level
CO1	Identify and demonstrate various safety equipment and procedures.	9	Understanding
CO 2	Perform risk assessment and hazard identification in industrial settings.	11	Analyzing
CO 3	Conduct accident investigations and develop safety reports.	11	Applying
CO 4	Implement emergency response plans and conduct safety drills.	10	Evaluating
	Lab Exam	4	

CO – PO Mapping:

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	2	1					

CO2		2	1				
CO3		1				1	2
CO4						2	

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline:

Module outcomes	Description	Duration (Hours)	Cognitive Level
CO1	Identify and demonstrate various safety equipment and procedures.		
M1.01	Study and demonstration of personal protective equipment (PPE)	4	Applying
M1.02	Use of gas detectors, fire alarms, and safety signage	5	Applying
CO2	Perform risk assessment and hazard identification in industrial settings.		
M2.01	Identifying hazards in industrial environments	5	Understanding
M2.02	Conducting risk assessment using standard methods	6	Applying
	Lab ExamI	2	
CO3	Conduct accident investigations and develop safety reports.		
M3.01	Case study analysis of industrial accidents	4	Understanding
M3.02	Steps in accident investigation and documentation	3	Applying
M3.03	Developing safety reports and corrective measures	4	Applying
CO4	Implement emergency response plans and conduct safety drills.		
M4.01	Study and execution of fire evacuation drills	3	Applying
M4.02	Conducting first aid and CPR training	3	Applying
M4.03	Emergency response planning and implementation	4	Analyzing
	Lab Exam – II	2	